CS203 · DIGITAL DESIGN PROJECT · FINAL DEMO

UART-Based Secure Digital Communication System

8-bit UART with ChaCha20 encryption and Hamming (13,8) error correction, written in Verilog

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INTRODUCTION

Project overview

A sender transmits variable-length messages to a receiver over a UART serial link. Each message is encrypted, protected against bit errors, sent serially, then corrected and decrypted at the receiver.

UART 8N1

8 data bits, no parity, 1 stop bit, idle-high line, LSB first. TX/RX FSMs with FIFO buffering and a baud generator.

ChaCha20

Stream cipher: a 256-bit key, 96-bit nonce and block counter make a keystream that is XORed with each byte.

Hamming (13,8)

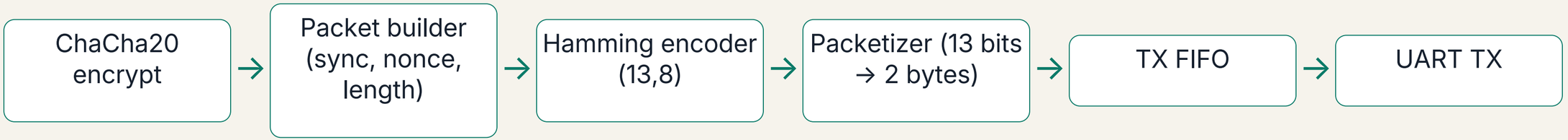
Extended Hamming code: corrects any single-bit error and detects double-bit errors in each codeword.

Scope: RTL and simulation (Icarus Verilog, VVP, GTKWave,Webpage).

ARCHITECTURE OVERVIEW

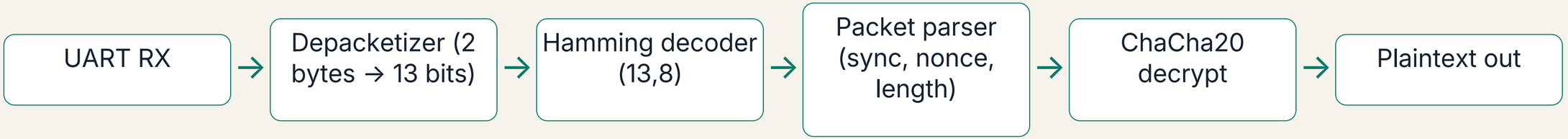
System architecture

TRANSMIT PATH



Serial TX/RX line between the two paths

RECEIVE PATH

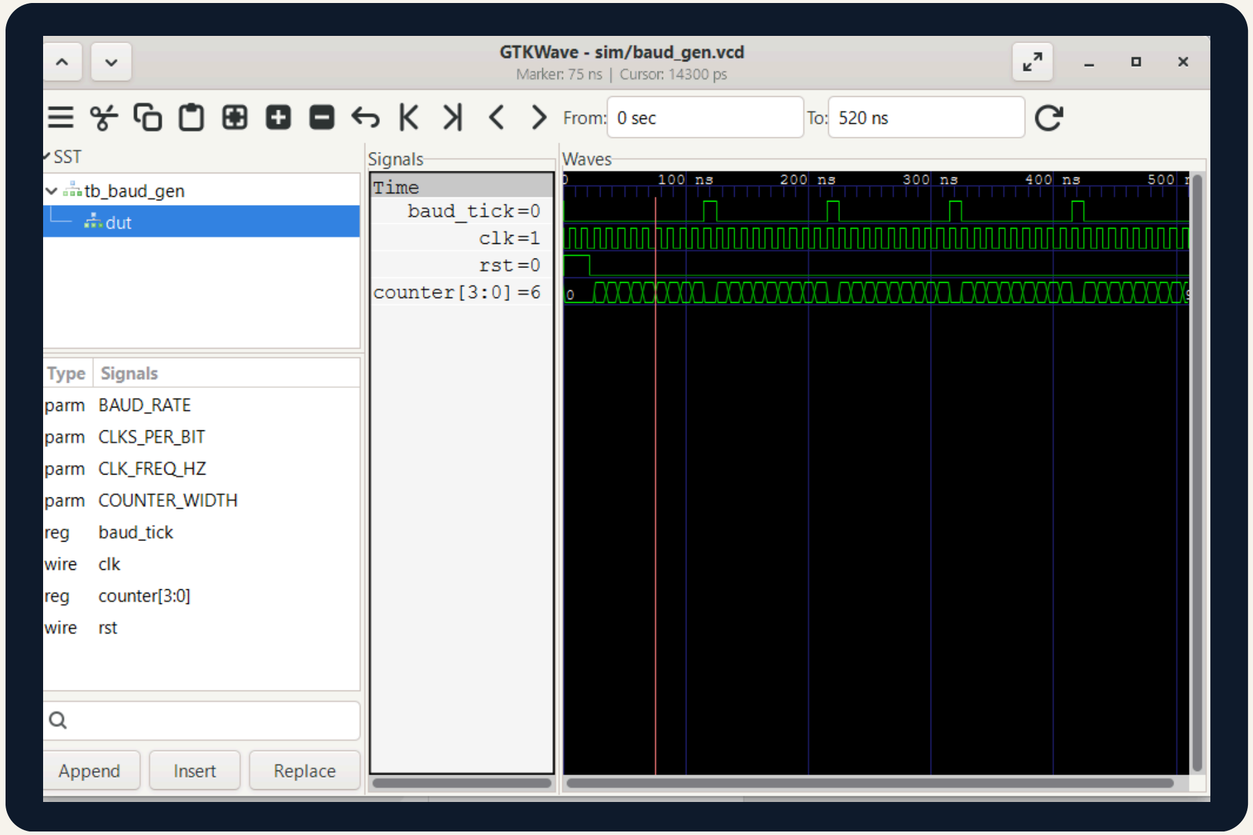


BLOCK: BAUD GENERATOR

Baud rate generator

A clock divider which generates baud ticks to synchronize the transmission of bits.

- CLKS_PER_BIT = CLK_FREQ_HZ / BAUD_RATE
- Default 50 MHz clock at 9600 baud: 5208 clocks per bit
- Parameterised, so any clock and baud rate can be used
- Testbench confirms tick spacing and reset behaviour (PASS)



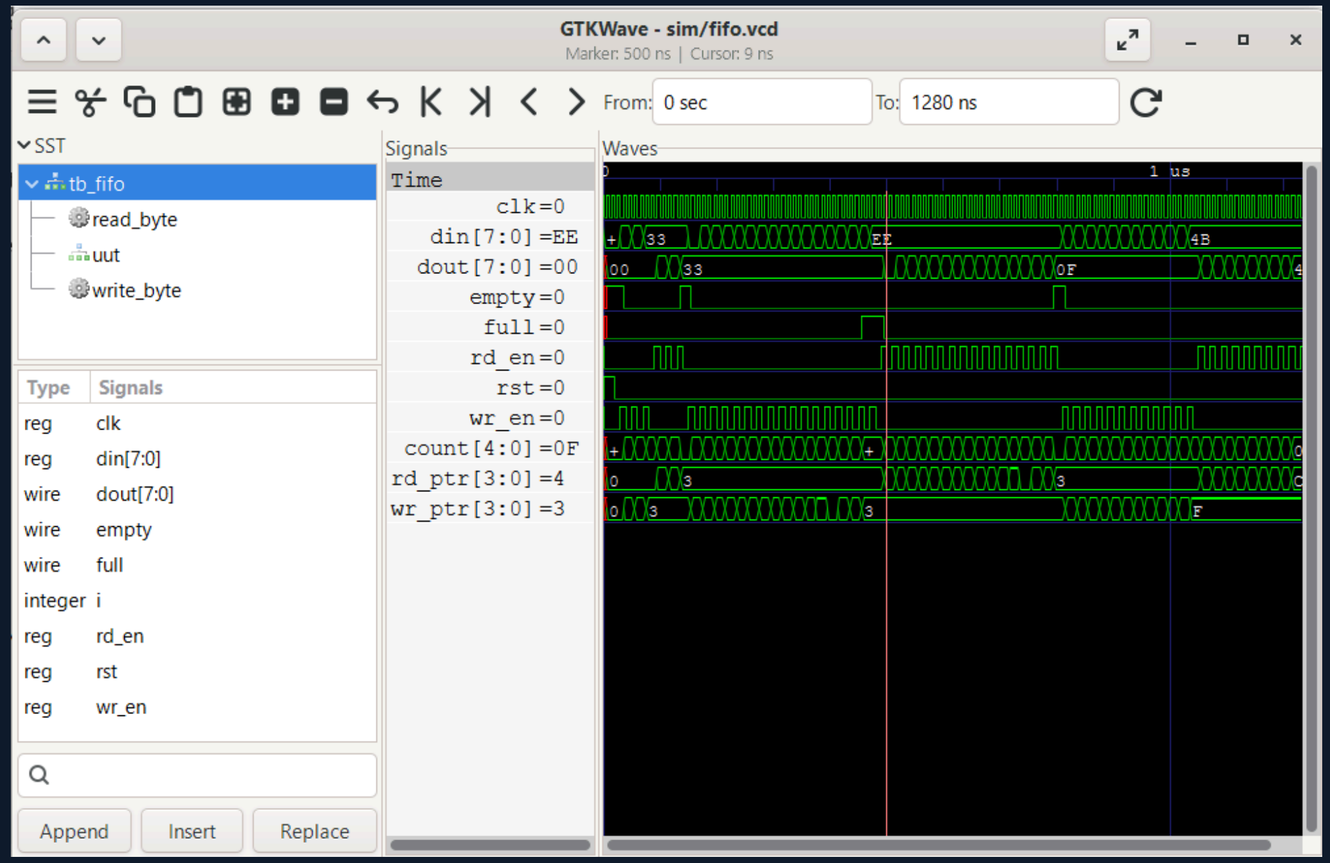
Baud generator simulation waveform

BLOCK: FIFO

FIFO buffer (16 × 8)

Buffers bytes ahead of the UART transmitter so the host is not stalled by serial speed.

- Memory array with read and write pointers
- Occupancy counter drives the full and empty flags
- Writes are rejected when full, reads when empty
- Ordering preserved, including simultaneous read and write (PASS)

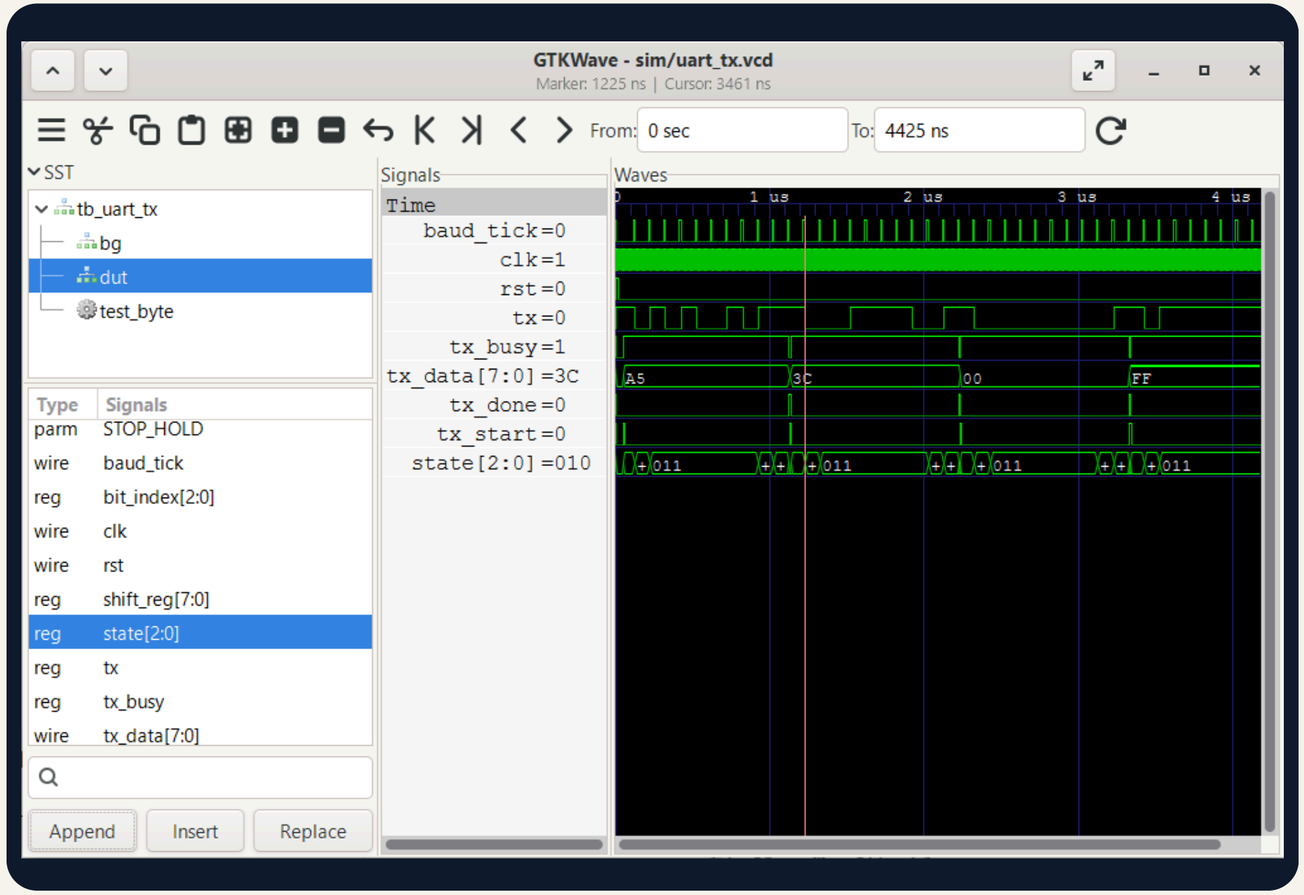


FIFO simulation waveform

UART transmitter

Serialises one byte into an 8N1 frame.

- FSM: IDLE, START_WAIT, START_BIT, DATA, STOP_BIT, STOP_HOLD
- Start bit low, 8 data bits LSB first, stop bit high
- Line idles high; busy and done status outputs
- Verified with 0xA5, 0x3C, 0x00 and 0xFF (PASS)



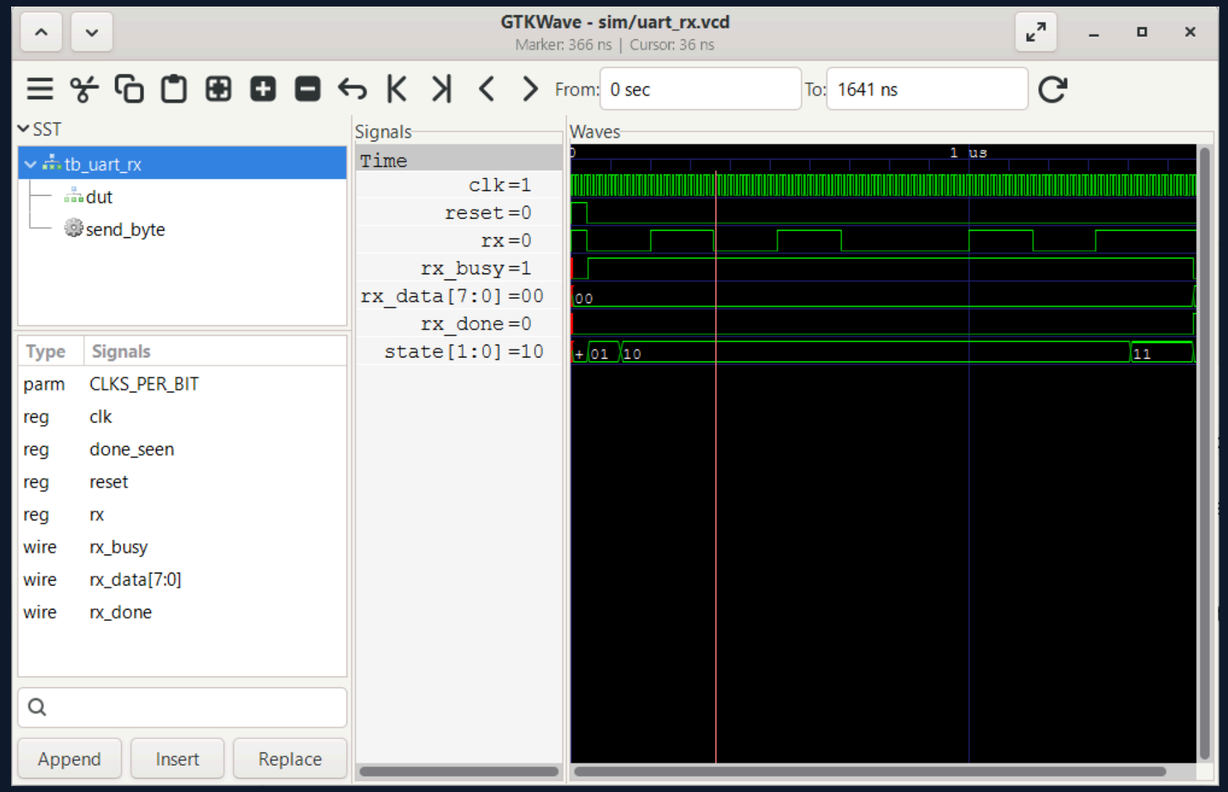
UART TX simulation waveform

BLOCK: UART RX

UART receiver

Detects a frame, samples each bit and rebuilds the byte.

- FSM: IDLE, START, DATA, STOP
- Incoming rx signal passes through rx_meta and rx_sync to reduce the risk of metastability
- Start bit re-checked at mid-bit to reject glitches
- Samples 8 data bits at bit centre, then checks the stop bit
- One-cycle rx_done pulse; 0xA5, 0x3C, 0x00, 0xFF received (PASS)



UART RX simulation waveform

SECURITY BLOCK

ChaCha20 encryption and decryption

Keystream generator

chacha20_keystream builds 64-byte blocks from a 256-bit key, a 96-bit nonce and a block counter. Shared by both directions.

Encrypt / decrypt

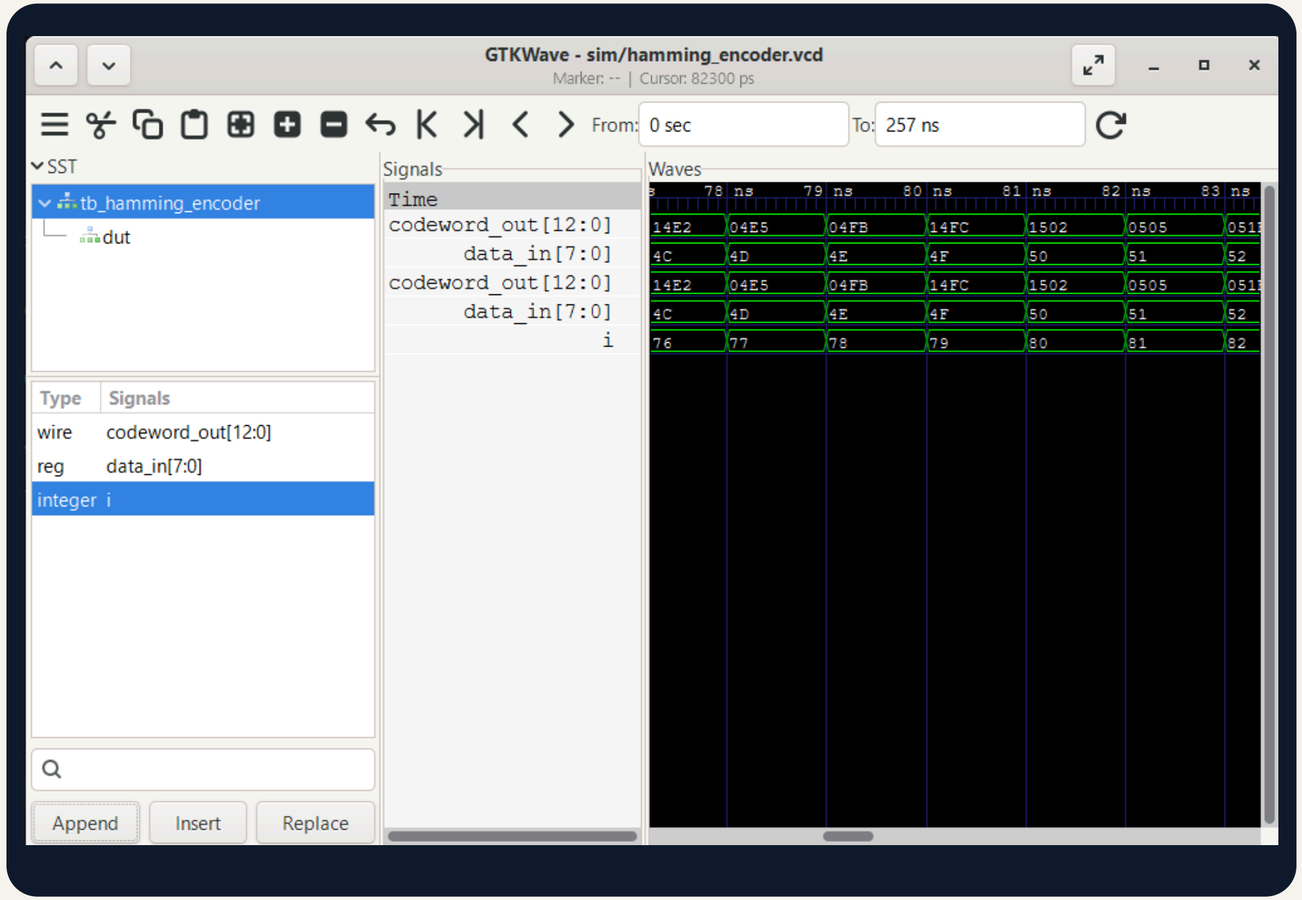
Each plaintext byte is XORed with the keystream. Decrypt applies the same keystream, so the same key, nonce and counter recover the plaintext.

- Nonce = {64-bit prefix, 32-bit counter}; counter stops rather than wrapping
- Tests: known ChaCha20 vector, 18-byte message, 70-byte message crossing the 64-byte block boundary (all PASS)

Hamming encoder and packetizer

hamming_encoder turns each 8-bit packet byte into a 13-bit codeword. hamming_packetizer then splits it into two UART bytes.

- Codeword: 8 data bits, 4 check bits (positions 0, 1, 3, 7) and 1 overall parity bit (bit 12)
- Encoder is combinational and was tested on all 256 inputs (PASS)
- Packetizer FSM: IDLE, SEND_LOW, SEND_HIGH, with byte_valid / byte_ready handshake
- First byte codeword[7:0], second byte {000, codeword[12:8]}



Hamming encoder simulation waveform

Hamming decoder and depacketizer

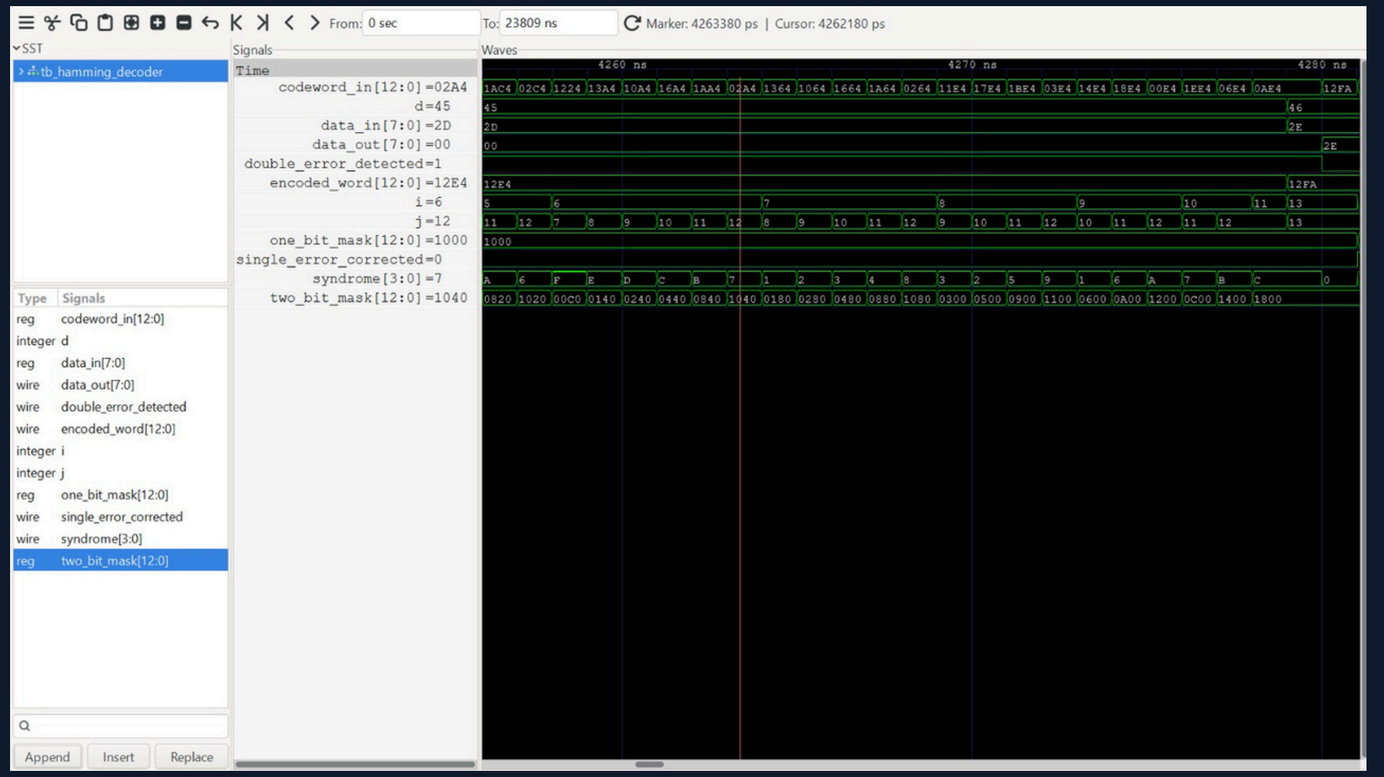
Depacketizer

hamming_depacketizer collects two UART bytes and rebuilds the 13-bit codeword. It raises codeword_valid when done and format_error if the 3 padding bits are not zero.

Decoder

hamming_decoder computes a syndrome from the codeword. A single-bit error is corrected (single_error_corrected). A double-bit error is flagged (double_error_detected) and reported upstream as a packet error.

Decoder test: all 256 values, all 13 single-bit error positions and all 78 double-bit pairs per value (PASS)
System test: encoder, packetizer, depacketizer and decoder chained, 6 cases including a corrected single error and a detected double error (PASS)



1INTEGRATION: secure_uart_top

Secure UART top level

One top-level module now wraps the whole system. The baud generator, FIFO, UART TX and UART RX built in Stage 2 sit inside it, together with the ChaCha20 and Hamming blocks.

Inside the top level

baud_gen, fifo (16 × 8), uart_tx, uart_rx, chacha20_keystream, chacha20_encrypt, chacha20_decrypt, hamming_encoder, hamming_packetizer, hamming_depacketizer, hamming_decoder.

Transmit interface

key and nonce_prefix inputs. Pulse tx_message_start when tx_message_ready is high, then stream tx_message_length bytes using tx_data_valid and tx_data_ready. Outputs: tx_message_done, tx_error.

Receive interface

rx_data with rx_data_valid, rx_data_last and rx_message_done. Status: rx_single_error_corrected and rx_packet_error. Serial pins: tx and rx.

Defaults: 50 MHz clock, 9600 baud, TX FIFO depth 16. Zero-length messages are rejected; lengths 1 to 65,535 bytes are supported by the format.

Packet format, nonce and control FSMs

Packet byte	Contents
0–1	Sync bytes 0xC3, 0x5A
2–13	96-bit nonce, MSB first
14–15	Message length (16 bits)
16 onward	ChaCha20 ciphertext

Every packet byte, header included, is Hamming-encoded before UART transmission. Nonce = {64-bit prefix, 32-bit counter}.

Control FSMs

- TX: IDLE, HEADER, PAYLOAD, WAIT_DRAIN
- RX: SYNC0, SYNC1, NONCE, LENGTH_HI, LENGTH_LO, PAYLOAD
- Secure loopback test passes for 5, 18 and 70-byte messages

SIMULATION

Verification results

Testbench	What it checks	Result
tb_baud_gen	Tick interval, reset	PASS
tb_fifo	Order, full/empty, underflow, simultaneous R/W	PASS
tb_uart_tx	4 patterns: A5, 3C, 00, FF	PASS
tb_uart_rx	4 patterns: A5, 3C, 00, FF	PASS
tb_chacha20	Known vector, 18-byte and 70-byte round trips	PASS
tb_hamming_encoder	All 256 inputs	PASS
tb_hamming_decoder	256 values × 13 single and 78 double errors	PASS
tb_hamming_system	6 cases through packetizer and depacketizer	PASS
tb_secure_uart_top	Loopback of 5, 18 and 70 bytes (93 bytes total)	PASS

TIMELINE

Weekly progress

Week	Dates	Work completed
Week 1	29 Sept – 5 Oct	Proposal, UART specification, tool setup (Icarus Verilog, VVP, GTKWave)
Week 2	5–12 Oct	Repository created. Baud generator, FIFO, UART TX, UART RX and basic top level written, each with a testbench. Waveforms captured. Stage 2 presentation.
Week 3	12–19 Oct	Refined TX, RX, FIFO, baud and top. Added ChaCha20 keystream, encrypt and decrypt, Hamming encoder, decoder, packetizer and depacketizer, secure top level and README.
Week 4	19-26 Oct	Testbenches, final testing, webpage, demo video and presentation.

RESOURCES

Project links

Webpage

https://secure-signal-lab--mishrarishikes3.replit.app/secure-transmission-demo/

Git repository

https://github.com/mishrarishikesh333/UART_Project

Demo video

https://drive.google.com/file/d/1o3tK29fiHzS65xmXHlXoVtSwy4bAPmGV/view?usp=sharing

References

Reference	Link	What it contributes
Cyphertalk ChaCha20	https://muens.io/chacha20/	This writeup explains how ChaCha20 works.
Hamming Code: An Analysis of its Reliability and Efficiency in Computer Networks Nawras H. Sabbry Alla Levina	https://ieeexplore.ieee.org/stamp/stamp.jsp?tp=&arnumber=10339080	Extends Hamming to single-error-correct, double-error-detect (SEC-DED) with an extra parity bit, as used in our (13,8) code.
Design and Implementation of UART SerialCommunication Module Based on FPGABiswajit Roy Dakua, Md. Ismail Hossain and Foisal Ahmed	Design and Implementation of UART Serial Communication Module Based on FPGA | Zenodo	Verilog UART core with baud generator and programmable control register; guided our modular TX/RX/baud split.

Contribution

Meet Trivedi
2025CSB1220

- Basic Top
- Hamming encoder
- Packetizer

Menkudle Shreyash
2025CSB1221

- Baud Generator
- Decryption

Mishra Rishikesh
2025CSB1223

- UART TX
- Encryption

Kanchi Aggarwal
2025CSB1204

- UART RX
- Hamming decoder
- Depacketizer

Gunjan Chaudhary
2025CSB1195

- FIFO Buffer
- Secure Top